Big Data Analytics

Previous question papers sorted unit-wise · Chaitanya Bharathi Institute of Technology

Tag used	Paper
Dec'24 CSE	R22 · 22ADE12 · B.E. CSE V Sem · Dec 2024
Nov'25 CSE	R22 · 22ADE12 · B.E. CSE V Sem · Nov 2025
May'26 IT/AI&DS	R22 · 22ADC12 · B.E. IT, AI&DS VI Sem · May 2026
Jun'24 IT	R20 · 20ADC14 · B.E. IT VI Sem · May-Jun 2024
Dec'23 CSE	R20 · 20CSE22 · B.E. CSE VII Sem · Nov-Dec 2023
Dec'23 AI&DS	R20 · 20ADC14 · B.E. AI&DS VII Sem · Nov-Dec 2023
Jun'23 IT	R20 · 20ADC14 · B.E. IT VI Sem · May-Jun 2023
Dec'22 IT	R18 · 18ITC27 · B.E. IT VII Sem · Dec 2022
Dec'22 CSE	R18 · 18CSE17 · B.E. CSE VII Sem · Dec 2022 (Data Science & BDA)

Unit	Topic	Questions
UNIT I	Introduction to Big Data	15
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UNIT III	Big Data Frameworks: MapReduce and YARN	22
UNIT IV	Hive and Pig	34
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Each question shows its paper, question number and marks. Similar questions are grouped under one topic; **IMP** marks topics asked three or more times, which are the safest to prepare first. Only questions within the syllabus are included; wording is kept as in the papers, with minor grammar fixes.

UNIT I · Introduction to Big Data

15 questions

Definition, importance and need for Big Data IMP · asked 6 times

1.	What is Big Data? Why is it important?	Jun'24 IT · Q1 · **3M**
2.	Define Big Data and list any two characteristics.	May'26 IT/AI&DS · Q1 · **2M**
3.	Briefly explain the importance of big data analytics.	Dec'22 CSE · Q1 · **2M**
4.	When to consider Big Data as a solution?	Dec'22 IT · Q1 · **2M**
5.	What is Big Data? Explain its role in identifying the patterns in Social Media.	Dec'22 IT · Q11a · **5M**
6.	Explain how Big Data differs from traditional data processing.	Nov'25 CSE · Q6a · **5M**

Characteristics of Big Data (3Vs / 4Vs / 5Vs) IMP · asked 4 times

1.	List and explain three V's of big data.	Jun'23 IT · Q1 · **3M**
2.	Recall the definitions of the basic three Vs.	Dec'23 CSE · Q1 · **3M**
3.	Illustrate and write the three characteristics of data.	Dec'23 CSE · Q7b · **4M**
4.	Illustrate the significance of Veracity in 4V's of big data.	Dec'22 CSE · Q7 · **2M**

Types and sources of data

1.	Illustrate Big Data types each with an example.	Dec'22 CSE · Q17b · **5M**
2.	Choose any 5 sources of big data and write the corresponding real-world use cases.	Dec'23 CSE · Q6a · **5M**

Applications of Big Data

1.	Describe the few use cases where Big Data is applicable.	Dec'22 IT · Q12a · **5M**
2.	Explain the need of big data for healthcare industry.	Dec'23 CSE · Q7a · **5M**

Data Science and Analytics

1.	Recall the similarities between Big Data Analytics and Data Science.	Dec'23 CSE · Q6b · **4M**

UNIT II · Hadoop Ecosystem

23 questions

Hadoop stack / ecosystem and its need IMP · asked 4 times

1. Define in brief about the Hadoop Stack for Big Data. — Dec'24 CSE · Q1 · **2M**
2. Explain the Big Data technology stack and the role of Hadoop in distributed data processing. — May'26 IT/AI&DS · Q6a · **5M**
3. Describe the architecture of the Hadoop stack with a neat diagram. — Nov'25 CSE · Q6b · **5M**
4. What is Hadoop Ecosystem? Why is this ecosystem needed when Hadoop MapReduce is already available? — Dec'23 CSE · Q8b · **4M**

Hadoop 1 vs Hadoop 2; comparison with SQL / traditional systems IMP · asked 3 times

1. Contrast the salient differences between Hadoop 1 and Hadoop 2. — Dec'23 CSE · Q2 · **3M**
2. Differentiate between the Hadoop file system and the traditional file system. — Nov'25 CSE · Q1 · **2M**
3. Classify between Big Data and SQL. — Dec'23 AI&DS · Q4 · **3M**

HDFS architecture and components IMP · asked 7 times

1. Define Hadoop File system. — Dec'23 AI&DS · Q1 · **3M**
2. Explain in detail with diagram about the architecture of HDFS. — Dec'24 CSE · Q6a · **5M**
3. Show the architecture of HDFS and explain the functions of NameNode and DataNode. — May'26 IT/AI&DS · Q6b · **5M**
4. Illustrate and explain the HDFS Architecture. — Dec'22 IT · Q11b · **5M**
5. Sketch and explain the HDFS architecture in detail. — Dec'22 CSE · Q18a · **5M**
6. What are the main components of big data HDFS? Explain in detail. — Dec'23 AI&DS · Q6a · **5M**
7. Explain HDFS Federation. — Dec'22 IT · Q2 · **2M**

HDFS block storage and replication

1. Explain in detail about the block replication architecture and replication method in HDFS. — Dec'24 CSE · Q7a · **5M**
2. Illustrate how HDFS stores large files using block storage and replication to ensure fault tolerance. — May'26 IT/AI&DS · Q7a · **5M**

HDFS operations: file read and write IMP · asked 4 times

1. Explain about file read and file write operations of HDFS. — Jun'24 IT · Q7 · **9M**
2. Discuss the anatomy of a FileRead and FileWrite in HDFS. — Dec'22 IT · Q12b · **5M**
3. List out the anatomy of a File Read and Write in HDFS. — Dec'23 AI&DS · Q6b · **4M**
4. How can a client read data from the HDFS? Explain with neat diagram. — Jun'23 IT · Q6a · **5M**

HDFS commands / Hadoop CLI IMP · asked 3 times

1. Illustrate the Command-Line Interface of Hadoop. Write the syntax of five commands used in Hadoop CLI. — Dec'24 CSE · Q7b · **5M**
2. List out HDFS file commands and explain with an example. — Dec'22 CSE · Q17a · **5M**
3. Illustrate touchz command with an example. — Dec'22 CSE · Q8 · **2M**

UNIT III · Big Data Frameworks: MapReduce and YARN

22 questions

MapReduce basics, architecture and working IMP · asked 8 times

1.	Define Map-Reduce.	Dec'22 IT · Q3 · **2M**
2.	Explain about the working of MapReduce. Give MapReduce example.	Dec'24 CSE · Q6b · **5M**
3.	Analyze the MapReduce execution process including map, shuffle, and reduce phases with an example.	May'26 IT/AI&DS · Q7b · **5M**
4.	What is MapReduce? Explain MapReduce architecture.	Jun'23 IT · Q7a · **5M**
5.	Illustrate and explain the Architecture of Map-Reduce.	Dec'22 IT · Q13a · **5M**
6.	Explain in detail about Map reduce with neat sketch.	Dec'22 CSE · Q19a · **5M**
7.	Design MapReduce program for word count and explain.	Nov'25 CSE · Q7a · **5M**
8.	Explain how cat command in MapReduce relates to SQL.	Dec'23 CSE · Q8a · **5M**

Developing and running MapReduce applications IMP · asked 6 times

1.	Design the steps in developing MapReduce application with an example.	Nov'25 CSE · Q7b · **5M**
2.	List the different Stages in developing a MapReduce Application.	Dec'22 IT · Q14b · **5M**
3.	Illustrate the steps to develop a Map Reduce application.	Dec'23 AI&DS · Q7b · **4M**
4.	How to run a Map Reduce program from a script file? Explain.	Dec'23 AI&DS · Q7a · **5M**
5.	Compare the stages of Job and Task in MapReduce.	Dec'22 IT · Q14a · **5M**
6.	List various failures in Map-Reduce.	Dec'22 IT · Q4 · **2M**

MapReduce types, I/O formats and input splits IMP · asked 4 times

1.	What is Map Reduce? Explain its types.	Jun'24 IT · Q6 · **9M**
2.	Write a short note on MapReduce Types and Formats.	Dec'22 IT · Q13b · **5M**
3.	Explain common Hadoop I/O formats.	Dec'23 CSE · Q12a · **5M**
4.	Define an input split. Illustrate how the input will split into records in MapReduce job.	Jun'23 IT · Q7b · **4M**

Joins, partitioner and pipelining IMP · asked 3 times

1.	Simplify the understanding of Map side join and Reduce side join.	Dec'23 CSE · Q12b · **4M**
2.	What are the key objectives of a partitioner in MapReduce?	Dec'23 CSE · Q13b · **4M**
3.	Explain the working of MapReduce Pipeline.	Dec'23 CSE · Q9a · **5M**

YARN

1.	Extend your understanding on resources management by YARN.	Dec'23 CSE · Q9b · **4M**

UNIT IV · Hive and Pig

34 questions

Hive: introduction, architecture and shell IMP · asked 6 times

1.	Define about querying data in Hive.	Dec'24 CSE · Q2 · **2M**
2.	Explain the purpose of HIVE Shell.	Dec'22 IT · Q8 · **2M**
3.	Construct HIVE architecture with a neat diagram.	Dec'23 AI&DS · Q8a · **5M**
4.	Describe the architecture of Hive with a neat diagram.	Nov'25 CSE · Q8a · **5M**
5.	Make use of the architecture of Hive and explain how HiveQL queries are executed in Hadoop.	May'26 IT/AI&DS · Q8b · **5M**
6.	Explain features and benefits of Hive in Hadoop.	Jun'23 IT · Q9b · **4M**

Hive data types, HiveQL, databases and commands IMP · asked 6 times

1.	Explain about the various data types supported by HIVEQL with an example.	Dec'22 IT · Q17b · **5M**
2.	Explain about HiveQL, user defined functions and aggregate function.	Dec'24 CSE · Q8b · **5M**
3.	Design a sample HiveQL script for querying student marks data.	Nov'25 CSE · Q9b · **7M**
4.	What are HIVE Commands? Explain.	Dec'23 AI&DS · Q9a · **5M**
5.	How can you create and manage the databases in Hive? Explain in detail.	Dec'22 IT · Q18b · **5M**
6.	Explain the creating, dropping and altering databases using Apache Hive.	Jun'23 IT · Q9a · **5M**

Hive UDFs

1.	Explain how user-defined functions enhance Hive query capabilities with examples.	Nov'25 CSE · Q9a · **3M**

Hive vs traditional databases IMP · asked 3 times

1.	Perform comparison of traditional database and Hive.	Dec'24 CSE · Q9a · **5M**
2.	Compare Hive with traditional databases.	Jun'24 IT · Q2 · **3M**
3.	Compare and contrast Hive and traditional relational database systems in handling large datasets.	May'26 IT/AI&DS · Q9b · **5M**

Pig: introduction, need and running Pig IMP · asked 7 times

1.	What is Apache Pig?	May'26 IT/AI&DS · Q2 · **2M**
2.	Demonstrate the necessity of PIG Latin.	Dec'23 AI&DS · Q8b · **4M**
3.	Explain how Pig can be used in practice for data processing.	Nov'25 CSE · Q8b · **5M**
4.	Explain about Pig, generating examples, and comparison with database.	Dec'24 CSE · Q9b · **5M**
5.	Compare pig with other databases.	Jun'23 IT · Q2 · **3M**
6.	Illustrate the steps of installing and running PIG.	Dec'23 AI&DS · Q9b · **4M**
7.	Describe two modes for running scripts in Pig.	Jun'23 IT · Q8a · **5M**

Pig data model and data types

1.	Explain the Pig data model including relations, tuples, and bags with examples.	May'26 IT/AI&DS · Q8a · **5M**
2.	Describe the various data types supported by pig in its data model with an example.	Dec'22 IT · Q18a · **5M**

Pig Latin, operators, scripts and UDFs **IMP · asked 8 times**

1.	Write the basic syntax of Pig-Latin Script.	Dec'22 IT · Q7 · **2M**
2.	Explain about pig Latin structure.	Jun'23 IT · Q8b · **4M**
3.	Explain about Pig Latin, user defined functions and data processing operators.	Dec'24 CSE · Q8a · **5M**
4.	Summarize the Data Processing Operators.	Dec'23 AI&DS · Q2 · **3M**
5.	What are user-defined functions in Pig?	Nov'25 CSE · Q2 · **2M**
6.	Discuss in brief about the scripting in PIG with suitable example.	Dec'22 IT · Q17a · **5M**
7.	Write a Pig Latin script to load a dataset and perform filtering and grouping operations.	May'26 IT/AI&DS · Q9a · **5M**
8.	Write the Pig Latin script to find the maximum temperature from the NCDC dataset.	Jun'24 IT · Q8 · **9M**

Pig vs Hive

1.	Compare Pig and Hive in detail.	Jun'24 IT · Q9 · **9M**

UNIT V · Apache Spark Framework

26 questions

Introduction to Spark; Spark vs MapReduce IMP · asked 5 times

1.	What is Apache Spark?	Dec'23 AI&DS · Q3 · **3M**
2.	What is Apache Spark? How is it different from Map Reduce?	Jun'24 IT · Q3 · **3M**
3.	Discover three key performance aspects when comparing Spark to MapReduce.	Dec'23 CSE · Q4 · **3M**
4.	Describe the present and the future of spark.	Jun'23 IT · Q3 · **3M**
5.	State the benefits and challenges of using Spark for big data processing.	Nov'25 CSE · Q11b · **5M**

Spark architecture and components IMP · asked 6 times

1.	What are the Components of SPARK?	Dec'22 IT · Q9 · **2M**
2.	Illustrate and explain the components of SPARK Framework.	Dec'22 IT · Q19a · **5M**
3.	Explain the components of Spark architecture in detail.	Jun'24 IT · Q10 · **9M**
4.	Show the basic architecture of Spark.	Dec'23 AI&DS · Q11a · **5M**
5.	Explain the architecture of Apache Spark, and explain fundamentals of Scala and functional programming.	May'26 IT/AI&DS · Q10a · **5M**
6.	Identify the role of driver program in Spark.	Dec'23 AI&DS · Q10a · **5M**

RDDs: concepts, significance and creation IMP · asked 9 times

1.	Explain Resilient Distributed Dataset (RDD).	May'26 IT/AI&DS · Q3 · **2M**
2.	Define spark concepts, resilient distributed dataset and creating RDDs.	Dec'24 CSE · Q11b · **5M**
3.	Explain the Resilient Distributed Dataset (RDD) with examples.	Dec'22 IT · Q20a · **5M**
4.	What is the significance of Resilient Distributed Datasets (RDD) in Spark?	Jun'23 IT · Q12b · **4M**
5.	What are the advantages of spark-Resilient Distributed Datasets?	Jun'23 IT · Q6b · **4M**
6.	How do the advantages of Spark RDDs contribute to improving big data processing?	Jun'23 IT · Q4 · **3M**
7.	Identify the properties and operations of RDDs in Spark and explain.	May'26 IT/AI&DS · Q10b · **5M**
8.	How to create RDD in the Apache Spark?	Jun'23 IT · Q12a · **5M**
9.	Explain in detail how RDDs are created from different data sources with examples.	Nov'25 CSE · Q10a · **5M**

Spark operations: transformations and actions IMP · asked 6 times

1.	What is the difference between transformations and actions in Spark?	Nov'25 CSE · Q3 · **2M**
2.	Explain about transformations and actions in the context of RDDs.	Jun'23 IT · Q13a · **5M**
3.	What is Spark Transformation? Explain.	Dec'23 AI&DS · Q13b · **4M**
4.	Compare and contrast operations and transformations in Spark with examples.	Nov'25 CSE · Q11a · **5M**
5.	Demonstrate how RDD transformations and actions are used to process data in Spark with an example.	May'26 IT/AI&DS · Q11a · **5M**
6.	What is RDD in Spark? Write a python program to implement various RDD transformations and actions.	Dec'23 CSE · Q13a · **5M**